

Diffusion Maps in Empirical Asset Pricing

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Modeling Asset Returns Through a Factor Model

Conditional latent factor model

$$r_{i,t} = \beta_{i,t-1}^\top f_t + \varepsilon_{i,t} \quad (1)$$

- $r_{i,t}$ excess return, $f_t \in \mathbb{R}^K$ latent factors, $\varepsilon_{i,t}$ idiosyncratic error

$$\beta_{i,t-1} = g(z_{i,t-1}), \quad z_{i,t-1} \in \mathbb{R}^P \quad (2)$$

- Factor exposures $\beta_{i,t-1}$ are a function of lagged characteristics $z_{i,t-1}$
- Learning g is equivalent to dimensionality reduction on the characteristic space: $\mathbb{R}^P \rightarrow \mathbb{R}^K$:

$$g : \mathbb{R}^P \longrightarrow \mathbb{R}^K, \quad K \ll P \quad (3)$$

Dimensionality Reduction and Learning β in Practice

- **PCA:**¹ static factor loadings β_i extracted from return covariance matrix; no characteristics
- **IPCA:**² $\beta_{i,t-1} = z_{i,t-1}^\top \Gamma$, linear map from characteristics to time-varying betas
- **Neural Networks:**³ $\beta_{i,t-1} = g(z_{i,t-1})$, nonlinear map via autoencoder architecture
- **Diffusion Maps (Baeder):**⁴ embed time-series distributions of returns via diffusion maps; estimate betas from rolling time-series regressions in embedding space

¹Bai & Ng (2002). Determining the Number of Factors in Approximate Factor Models. *Econometrica*.

²Kelly, Pruitt & Su (2019). Characteristics Are Covariances. *JFE*.

³Gu, Kelly & Xiu (2021). Autoencoder Asset Pricing Models. *Journal of Econometrics*.

⁴Baeder, M.A. (2022). Manifold Learning for Empirical Asset Pricing. M.S. Thesis, Johns Hopkins University. Advisor: B. Sadiq.

Our Approach: Diffusion Maps on Characteristic Space

$$\tilde{Z}_{t-1} \in (-1, 1)^{N_t \times P} \xrightarrow{\Phi_t} B_{t-1} \in \mathbb{R}^{N_t \times K}, \quad K = 5, P = 94 \quad (4)$$

- Rank-normalized characteristic matrix compressed to K -dimensional diffusion coordinates
- Data: CRSP monthly US equity returns and 94 firm characteristics, 1990–2018

Approach 1: Risk-Premia (Embed Assets)

$$\tilde{z}_{i,t-1} \in \mathbb{R}^P \mapsto \beta_{i,t-1} \in \mathbb{R}^K \quad (5)$$

- Each of N_t stocks is a point in $\mathbb{R}^P$
- Embedding gives betas; estimate f_t via ridge regression on returns

Approach 2: Clustering (Embed Months)

$$\bar{x}_\tau = \text{std}_i(\tilde{z}_{i,\tau}) \in \mathbb{R}^P \mapsto \phi_\tau \in \mathbb{R}^K \quad (6)$$

- Each of $M=60$ rolling months is a point in $\mathbb{R}^P$
- Find nearest neighbor months in diffusion space; forecast returns by softmax-weighted average

Data

- **Universe:** $P = 94$ firm-level **characteristics** (size, value, momentum, idiosyncratic volatility, turnover, etc.), sourced from the Open Source Asset Pricing data library
- **Returns:** US equities from CRSP, monthly frequency, 1990–2018, $r_{i,t}$ is the monthly excess return for stock i
- **Pre-processing:** rank-normalize each characteristic cross-sectionally each month to $(-1, 1)$:

$$\tilde{z}_{i,t-1}^{(p)} = \frac{2 \cdot \text{rank}(z_{i,t-1}^{(p)})}{N_t + 1} - 1 \quad (7)$$

- Minimum $N_t \geq 200$ stocks required per month
- Evaluation window for strategy validation: March 1997 – October 2018 (182 months)

Characteristic Embedding: Kernel and Markov Operator

Given rank-normalized cross-section $\tilde{Z}_{t-1}$, stocks are points in $\mathbb{R}^P$.

Step 1: Gaussian kernel (pairwise similarity)

$$W_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{\epsilon}\right), \quad x_i = \tilde{z}_{i,t-1} \quad (8)$$

Step 2: Density normalization (Coifman–Lafon, $\alpha = 0.5$)⁵

$$\hat{W}_{ij} = \frac{W_{ij}}{d_i^\alpha d_j^\alpha}, \quad d_i = \sum_j W_{ij} \quad (9)$$

Step 3: Row-normalize to Markov matrix

$$P_{ij} = \frac{\hat{W}_{ij}}{\hat{d}_i}, \quad \hat{d}_i = \sum_j \hat{W}_{ij} \quad (10)$$

P is a row-stochastic matrix whose (i, j) entry gives one-step transition probability from stock i to stock j in a random walk over characteristic similarity graph.

⁵Coifman, R.R. and Lafon, S. (2006). Diffusion Maps. *Applied and Computational Harmonic Analysis*, 21(1), 5–30.

Characteristic Embedding: Diffusion Coordinates

Step 4: Eigendecomposition of P

$$P\psi_k = \lambda_k\psi_k, \quad 1 = \lambda_0 \geq |\lambda_1| \geq |\lambda_2| \geq \dots \quad (11)$$

Step 5: Diffusion map embedding (retain K nontrivial eigenvectors)

$$\Phi(x_i) = (\lambda_1\psi_1(x_i), \lambda_2\psi_2(x_i), \dots, \lambda_K\psi_K(x_i)) \in \mathbb{R}^K \quad (12)$$

Diffusion distance: the Euclidean distance in diffusion coordinates integrates over all paths of length t on the graph:

$$D_t(x_i, x_j)^2 = \sum_{k=1}^K \lambda_k^{2t} (\psi_k(x_i) - \psi_k(x_j))^2 \quad (13)$$

- More robust to noise than geodesic distance
- Captures intrinsic geometry of the characteristic manifold
- Output: $B_{t-1} = \Phi(\tilde{Z}_{t-1}) \in \mathbb{R}^{N_t \times K}$, the beta matrix for month t

Risk-Premia Approach: Factor Estimation and Prediction

Each month t , embed N_t stocks in characteristic space to obtain $B_{t-1} \in \mathbb{R}^{N_t \times K}$.

Step 1: Estimate latent factors via ridge regression

$$\hat{f}_t = \arg \min_{f \in \mathbb{R}^K} \|r_t - B_{t-1}f\|^2 + \lambda \|f\|^2 = (B_{t-1}^\top B_{t-1} + \lambda I)^{-1} B_{t-1}^\top r_t \quad (14)$$

with $\lambda = 10^{-6}$ (regularization).

Step 2: Expanding-window average risk premia

$$\hat{\lambda}_t = \frac{1}{t-1} \sum_{s=1}^{t-1} \hat{f}_s \in \mathbb{R}^K \quad (15)$$

Step 3: Out-of-sample predicted return

$$\hat{r}_{i,t}^{\text{pred}} = \beta_{i,t-1}^\top \hat{\lambda}_t, \quad \beta_{i,t-1} = \Phi(\tilde{z}_{i,t-1}) \quad (16)$$

- Prediction uses only information available before month t
- Analogous to Fama–MacBeth second step, but with nonlinear (diffusion map) betas⁶

⁶Fama, E.F. and MacBeth, J.D. (1973). Risk, Return, and Equilibrium: Empirical Tests. *Journal of Political Economy*, 81(3), 607–636.

Clustering Approach: Month Embedding and Neighbor Forecast

Step 1: Month-level feature vector — cross-sectional std of rank-normalized characteristics:

$$\bar{x}_\tau^{(p)} = \text{std}_i \left(\tilde{z}_{i,\tau}^{(p)} \right) \in \mathbb{R}^P \quad (17)$$

Step 2: Diffusion map on rolling window of $M = 60$ past feature vectors

$$X_{\text{past}} = [\bar{x}_{t-M}, \dots, \bar{x}_{t-1}] \in \mathbb{R}^{M \times P} \xrightarrow{\Phi_t} Z_{\text{past}} \in \mathbb{R}^{M \times K} \quad (18)$$

Current month $\bar{x}_t$ is then out-of-sample transformed: $\phi_t = \Phi_t(\bar{x}_t) \in \mathbb{R}^K$

Step 3: k -nearest neighbors and softmax weights

$$d_\tau^2 = \|\phi_t - Z_{\text{past},\tau}\|^2, \quad w_\tau \propto \exp(-d_\tau^2 / T), \quad \tau \in \mathcal{N}_k(t) \quad (19)$$

Step 4: Predicted return for stock i in month t

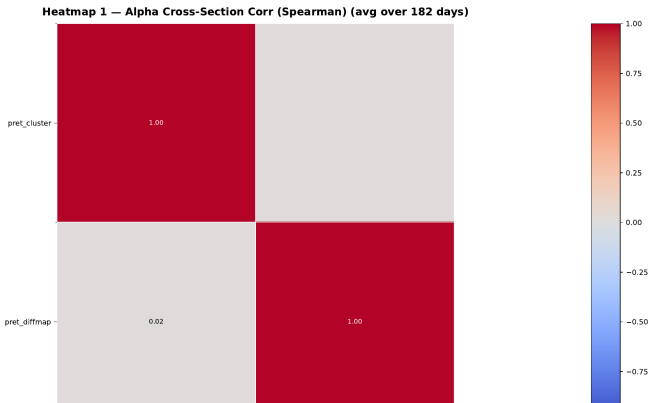
$$\hat{r}_{i,t} = \sum_{\tau \in \mathcal{N}_k(t)} w_\tau r_{i,\tau} \quad (20)$$

- $k = 10$ nearest neighbors, $K = 5$ diffusion dimensions, rolling window $M = 60$ months

Strategy Validation: Alphamark Framework

- **Alphamark** is a strategy validation pipeline that we are using to evaluate predicted return signals $\hat{r}_{i,t}$ against realized returns $r_{i,t}$
- Stocks are sorted into quantile portfolios (qr_100, qr_75, qr_50, qr_25) of equal bet sizing from predicted return each month
- **Key initial finding:** the two signals ($\hat{r}^{\text{cluster}}$ and $\hat{r}^{\text{diffmap}}$) are nearly uncorrelated (cross-sectional Spearman correlation ≈ 0.02)

Window: 1997-03-31 → 2018-10-31 | Days: 182



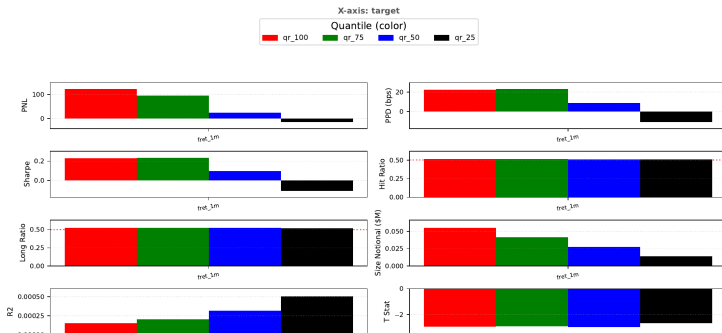
Results: Risk-Premia Approach

Signal: $= \hat{r}_{i,t}^{\text{diffmap}}$

- Positive cumulative PnL at qr_100 and qr_75 (≈ 125 and ≈ 90 respectively)
- Sharpe ratio ≈ 0.2 at qr_100/qr_75; negative at qr_25 and qr_50
- $R^2 \approx 1.5 \times 10^{-4}$ at qr_50/qr_25, T-stat *negative* (≈ -2) — signal partially inverted at lower quantiles
- Cumulative PPD positive at qr_100/qr_75 ($\approx 4,000$ bps); negative at qr_25 ($\approx -6,000$ bps)

Window: 1997-03-31 → 2018-10-31 | Days: 182

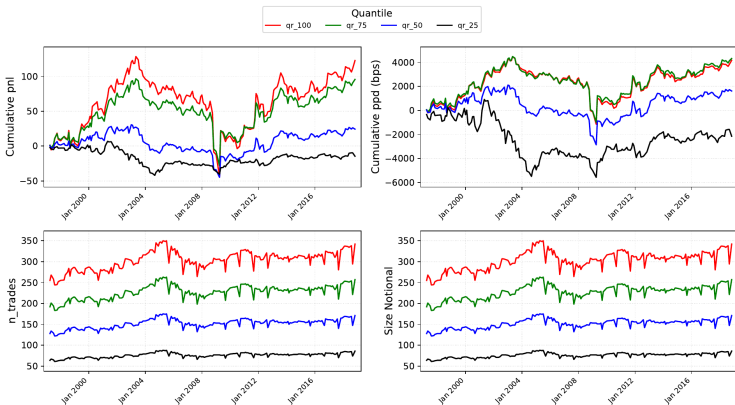
Bar Plots | signal: pret_diffmap bet_size_col: betsize_equal



Results: Risk-Premia Approach

fret_1m | pret_diffmap | betsize_equal — Temporal metrics

Window: 1997-03-31 → 2018-10-31 | Days: 182



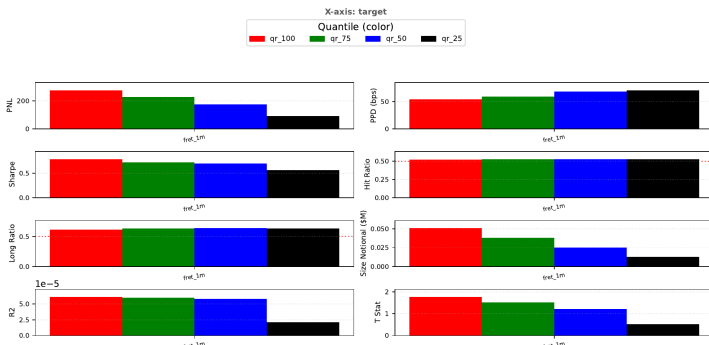
Results: Clustering Approach

Signal: $= \hat{r}_{i,t}^{\text{cluster}}$

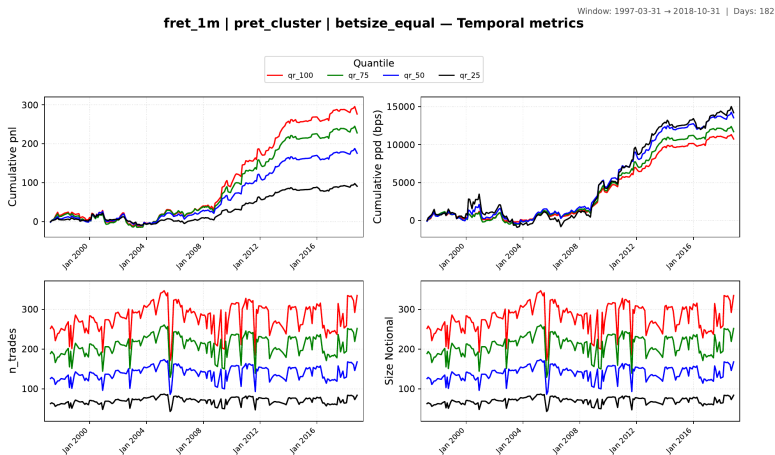
- Positive cumulative PnL across all quantile portfolios (qr_100 $\approx$ 300 by end of sample)
- Cumulative PPD reaches $\approx$ 15,000 bps at qr_100 over the full window
- Sharpe ratios positive across quantiles ($\approx$ 0.7 at qr_100)
- $R^2 \approx 5 \times 10^{-5}$, T-stat $\approx$ 1.5–2.0 at qr_100
- Strongest performance post-2008; PnL growth accelerates from 2010 onward

Bar Plots | signal: pret_cluster bet_size_col: betsize_equal

Window: 1997-03-31 → 2018-10-31 | Days: 182



Results: Clustering Approach



Future Methodology

- **Vector Diffusion Maps (VDM):**⁷ augment each graph edge with an orthogonal transformation O_{ij} encoding local tangent space alignment; affinity defined by consistency of transformations along paths, generalizing DM from the Laplace-Beltrami operator (scalars) to the connection-Laplacian (vector fields)
- **Rolling vs. expanding distribution:** the current risk-premia approach uses an expanding window for $\hat{\lambda}_t$; a rolling window may better capture regime changes in factor risk premia
- **Joint embedding:** embed both stocks and months simultaneously to allow information sharing across the two approaches
- **Kernel bandwidth selection:** systematic cross-validation of ϵ and k -neighbor parameters; current implementation uses library defaults
- **Out-of-sample R^2 :** formal computation of pooled predictive R^2 across the full evaluation window for both approaches, as in Baeder (2022)

⁷Singer, A. and Wu, H.-T. (2011). Vector Diffusion Maps and the Connection Laplacian. *arXiv:1102.0075*.

Collaborators and Current Status

- **Other collaborator:** Griffin Pinney — joint development of methodology, data pipeline, and implementation
- **Current status:** Work in progress. The two embedding approaches are implemented and validated through Alphamark. Results are preliminary — no definitive conclusions about outperformance over linear baselines (IPCA, PCA) have been established yet.
- **Data:** CRSP (returns) and Open Source Asset Pricing (characteristics, 94 signals, 1990–2018)

Summary

- **Motivation:** firm characteristics determine exposure to latent market factors; learning the intrinsic geometry of characteristics may improve return prediction
- **Approach 1 (Risk-Premia):** diffusion map on N_t stocks each month → nonlinear beta matrix → ridge regression for factors → expanding-average risk premia forecast
- **Approach 2 (Clustering):** diffusion map on 60-month rolling window of market states → nearest-neighbor months → softmax-weighted return forecast
- **Findings:** the two signals are nearly orthogonal (cross-sectional correlation ≈ 0.02); clustering approach shows more consistent positive PnL; risk-premia approach captures extreme quantile behavior but is more volatile
- **Caveats:** no formal R^2 benchmarking against linear models yet; results are preliminary and highly dependent on hyperparameter choices
- **Next steps:** VDMS, rolling distribution, joint signal combination, formal out-of-sample evaluation